ESP32-S3 Mouse Keep-Alive Firmware

Setup & Usage Guide Overview

Mouse Keep-Alive keeps your computer awake by simulating real USB mouse movement. It prevents idle timeouts in massaging apps, and other applications — fully configurable through a built-in web interface.

Note: This is firmware only. You must provide your own ESP32-S3 device.

When powered for the first time (or if no Wi-Fi credentials are saved), the device automatically starts in **Access Point (AP) mode** so you can configure it.

⚡ Quick Setup (30 Seconds)

1. Plug the ESP32-S3 into USB power.
2. Connect to the Wi-Fi network: **"The Micromaker Mouse Keep-Alive"**
3. Open a browser and go to: http://192.168.4.1
4. Select your Wi-Fi network (or enter it manually), type your password, and press **Connect**.
5. The device will reboot and connect to your Wi-Fi automatically.

After setup, access the web interface at: http://mouse.local
Done ✅

📶 Detailed Setup Instructions

Step 1 — Power the Device

Connect the ESP32-S3 to USB power.
If no Wi-Fi credentials are stored, it will create its own temporary Wi-Fi network (Access Point mode).

Step 2 — Connect to the Setup Network

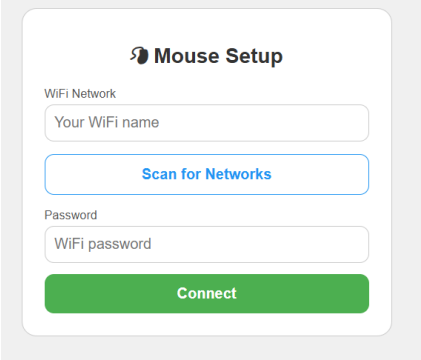
On your computer or smartphone:

1. Open Wi-Fi settings
2. Find and select: **The Micromaker Mouse Keep-Alive**
3. Connect to it

Note: Internet access is not required during setup.

Step 3 — Open the Configuration Page

After connecting to the device:

- Open your web browser
- Enter the following address: http://192.168.4.1
  The Wi-Fi configuration page will load.
- 

Step 4 — Enter Your Wi-Fi Credentials

You can:

- Click **Scan for Networks** and select your Wi-Fi
  OR
- Manually enter your Wi-Fi name (SSID)

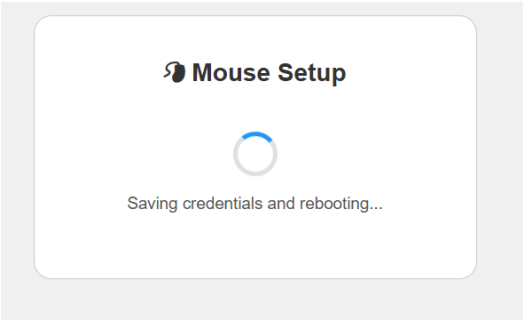
Then:

- Enter your Wi-Fi password
- Click **Connect**

Step 5 — Automatic Reboot

The device will:

1. Save your Wi-Fi credentials
2. Restart automatically
3. Connect to your home or office Wi-Fi
4. Disconnect USB cable wait for 5 seconds and Reconnect it.



Once Wi-Fi is connected, disconnect USB cable wait for 5 seconds and reconnect it, you can now access the web interface at: http://mouse.local
If that does not work, check your router's connected devices list to find the assigned IP address.

🔄 Resetting Wi-Fi

If you erase the saved credentials, move to another location or reflash the firmware, the device will automatically return to AP setup mode.

How to Use the Mouse Keep-Alive

1️⃣ Connect to Your Computer

Plug the ESP32-S3 into your computer using a USB cable.
The device will be detected automatically as a standard USB mouse (no drivers required).

2️⃣ Open the Web Interface

Make sure the device is connected to your Wi-Fi network.
In your browser, open: http://mouse.local
(If that doesn't work, check your router for the device IP address.)

3️⃣ Configure Movement Settings

Inside the web interface you can:

- Enable / Disable mouse movement
- Set movement interval (how often it moves)
- Adjust movement distance
- Enable random movement patterns
- Configure timing behavior

Press **Save** after making changes.

4️⃣ Let It Run

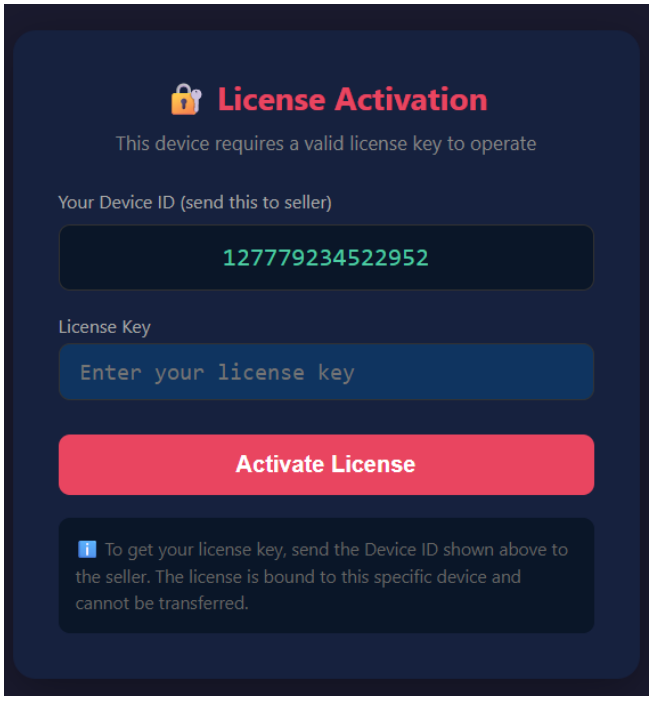
Once enabled, the device will:

- Simulate small real mouse movements
- Prevent idle timeouts
- Keep your computer status active

It runs silently in the background.

🔒 License Activation

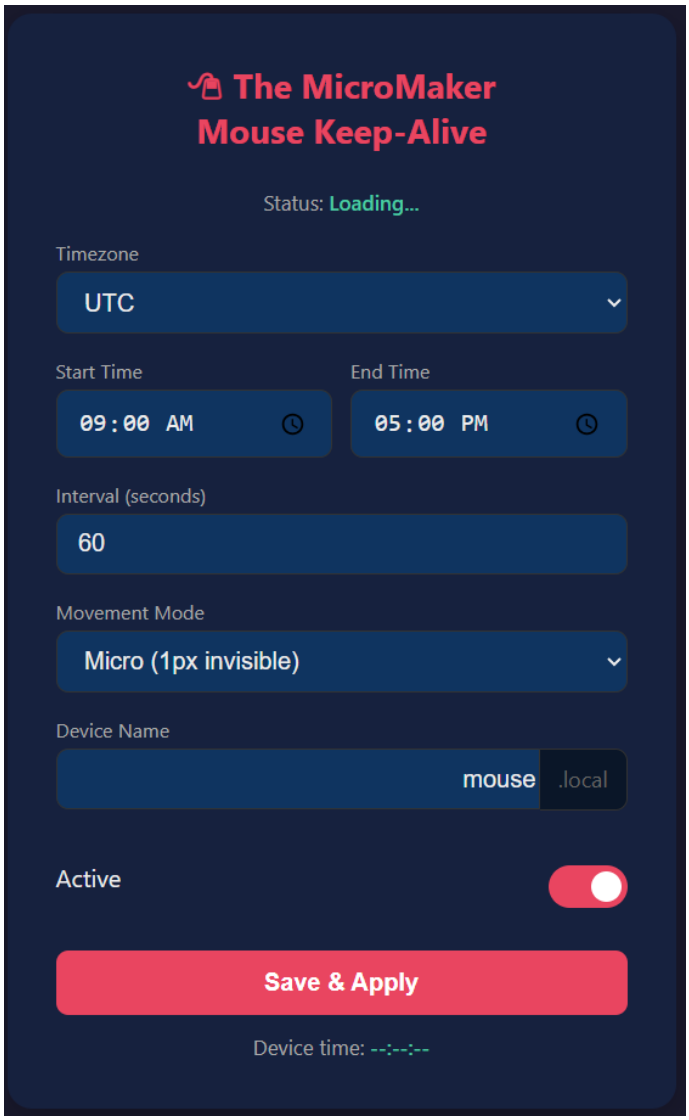
1. On first access, the web interface will show your **Device ID (Chip ID)**.
2. **After purchasing a license**, email the Device ID to **themicromaker@outlook.com** to receive your activation key.
3. Enter the key into the web interface and click **Activate**.
4. The device will verify the license and restart. Once activated, the firmware is fully functional.



⚠️ Important Notes

- The movement is small and designed not to disturb your work.
- You can disable it anytime from the web interface.

- Unplugging the USB cable immediately stops the device.



If you have any questions, contact: **themicromaker@outlook.com**